

Ganesh Gupta

817-822-1131 | iamgs10rk@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Owned ML development from problem formulation through **offline evaluation**, **A/B testing**, and **production rollout**, defining quality and regression metrics that automatically blocked underperforming model releases.
- Implemented **GraphRAG** within a production **retrieval and ranking** pipeline, combining **LLM-derived** semantic signals with graph context to improve relevance on complex, multi-hop queries.
- Built **distributed search** and retrieval infrastructure using **Elasticsearch** and knowledge-graph enrichment, engineering a pluggable pipeline executor via typed **REST APIs** at sub-200ms p99 latency.
- Built a deployment-gated **evaluation framework** enforcing guardrails and quality/regression checks across model outputs, blocking **15%** of failed releases before reaching production.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Owned an end-to-end **ranking and retrieval** system, redesigning **candidate generation** with **hybrid dense-sparse retrieval** and **re-ranking**; improved **Recall@3** from **61% to 89%** across a 2K-query **offline evaluation** set.
- Designed a **Neo4j** entity-relation graph for candidate expansion and multi-hop relevance signals in a production retrieval pipeline serving **50K+ daily requests**.
- Cut **p99 latency 420ms to 140ms** across **50K+ daily requests** by redesigning synchronous serving into an **async**, autoscaled architecture on **GCP Cloud Run**.
- Built **PySpark** data processing jobs preparing large-scale feature datasets for downstream model training and evaluation pipelines.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Architected and operated **25+ production Java** microservices serving **1M+ users** at **99.9% uptime**, owning system design, deployment, monitoring, and failure recovery across distributed services.
- Standardized failure-handling contracts across 25+ backend services, reducing production escalations **60%** in one quarter through consistent retry, timeout, and error-propagation semantics.

PROJECTS

LangFetch

tinyurl.com/LangFetch

- Built a multi-stage retrieval pipeline combining **Pinecone-based candidate generation**, **hybrid retrieval**, and relevance **re-ranking**, with **LangGraph** query decomposition for complex information needs.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Built an ML **evaluation platform** with **rubric-based scoring**, regression detection, hallucination analysis, and drift tracking across model versions, converting subjective quality criteria into measurable release metrics.

Distributed KV Store

github.com/ganeshhgupta/distributed-kv-store

- Built a **Raft-based** distributed KV store supporting concurrent writes, **linearizable reads**, failure recovery, and **log compaction**.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, Java, C++, TypeScript, SQL

Ranking & Retrieval: Candidate Generation, Hybrid Dense-Sparse Retrieval, Embedding Retrieval, Re-Ranking, Vector Search (Pinecone), GraphRAG

ML & Experimentation: PyTorch, Recall@K, Offline Evaluation, A/B Testing, Metric Design

Production ML: End-to-End Model Ownership, Evaluation Frameworks, Model Rollout, Observability, Low-Latency Model Serving

Knowledge Graphs: Neo4j, Entity-Relation Modeling, Graph-Based Retrieval